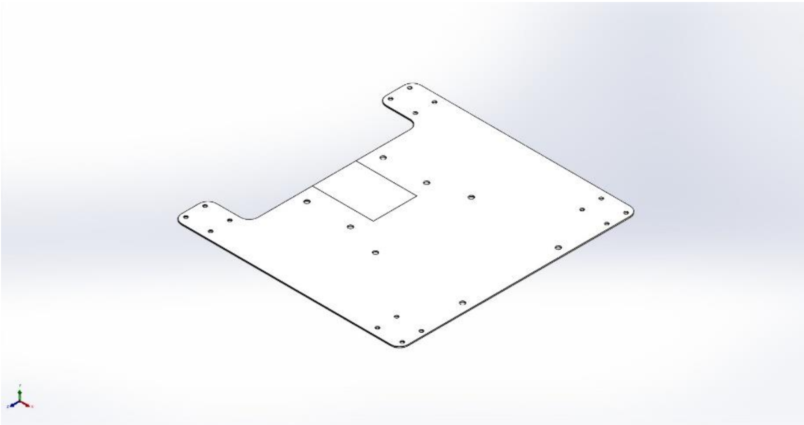


Simulation of Base Plate

Date: Monday, December 8, 2025
Designer: Solidworks
Study name: Static 1
Analysis type: Static

Table of Contents

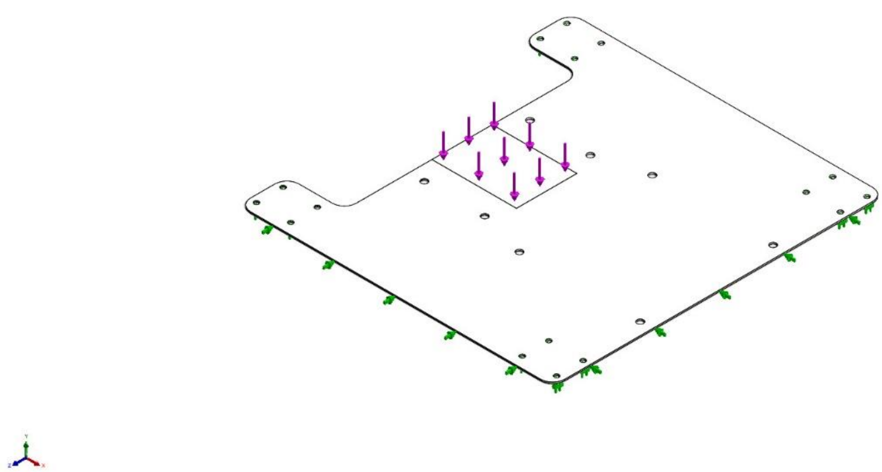
Description 1
Assumptions.. 2
Model Information 2
Study Properties................................... 3
Units 3
Material Properties 4
Loads and Fixtures 5
Connector Definitions 6
Interaction Information 6
Mesh information.................................. 7
Sensor Details....................................... 8
Resultant Forces 9
Beams 9
Study Results 10
Conclusion.. 12

Description

No Data

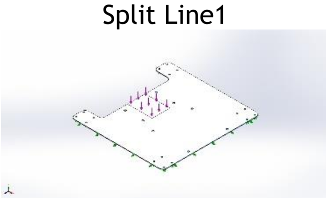
Assumptions

Model Information



Model name: Base Plate
Current Configuration: Default<As Machined>

Solid Bodies

Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Split Line1 	Solid Body	**Mass:1.24953 kg** **Volume:0.000158168 m^3** **Density:7,900 kg/m^3** **Weight:12.2454 N**	**C:\Users\masonbuehler\Downloads\CAD-20251208T223626Z-1-001\CAD\Tilty Loader\Base Plate.SLDPRT** Dec 8 17:25:41 2025

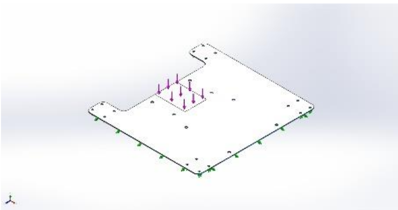
Study Properties

Study name	Static 1
Analysis type	Static
Mesh type	Solid Mesh
Thermal Effect:	On
Thermal option	Include temperature loads
Zero strain temperature	298 Kelvin
Include fluid pressure effects from SOLIDWORKS Flow Simulation	Off
Solver type	Automatic
Inplane Effect:	Off
Soft Spring:	Off
Inertial Relief:	Off
Incompatible bonding options	Automatic
Contact penalty stiffness scale factor	1
Large displacement	Off
Compute free body forces	On
Friction	Off
Use Adaptive Method:	Off
Result folder	SOLIDWORKS document (C:\Users\masonbuehler\Downloads\CAD-20251208T223626Z-1-001\CAD\Tilty Loader)

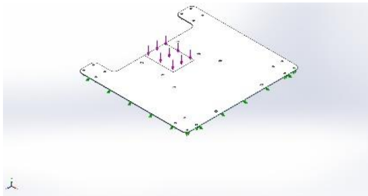
Units

Unit system:	SI (MKS)
Length/Displacement	mm
Temperature	Kelvin
Angular velocity	Rad/sec
Pressure/Stress	N/m^2

Material Properties

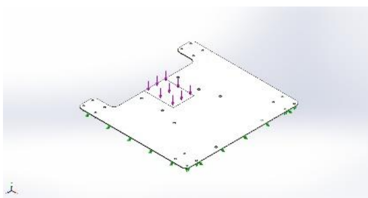
Model Reference	Properties	Components
	**Name:** AISI 1020 **Model type:** Linear Elastic Isotropic **Default failure criterion:** Unknown **Yield strength:** 3.51571e+08 N/m^2 **Tensile strength:** 4.20507e+08 N/m^2 **Elastic modulus:** 2e+11 N/m^2 **Poisson's ratio:** 0.29 **Mass density:** 7,900 kg/m^3 **Shear modulus:** 7.7e+10 N/m^2 **Thermal expansion coefficient:** 1.5e-05 /Kelvin	SolidBody 1(Split Line1)(Base Plate)
Curve Data:N/A		

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Roller/Slider-1		**Entities:** 2 face(s) **Type:** Roller/Slider

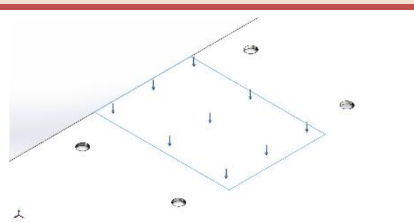
Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	-6.10352e-05	0	6.8903e-05	9.20484e-05
Reaction Moment(N.m)	0	0	0	0

Fixture name	Fixture Image	Fixture Details
Roller/Slider-2		**Entities:** 16 face(s) **Type:** Roller/Slider

Resultant Forces

Components	X	Y	Z	Resultant
Reaction force(N)	0	222.4	0	222.4
Reaction Moment(N.m)	0	0	0	0

Load name	Load Image	Load Details
Force-1		**Entities:** 1 face(s) **Type:** Apply normal force **Value:** 222.4 N

Connector Definitions
No Data

Interaction Information
No Data

Mesh information

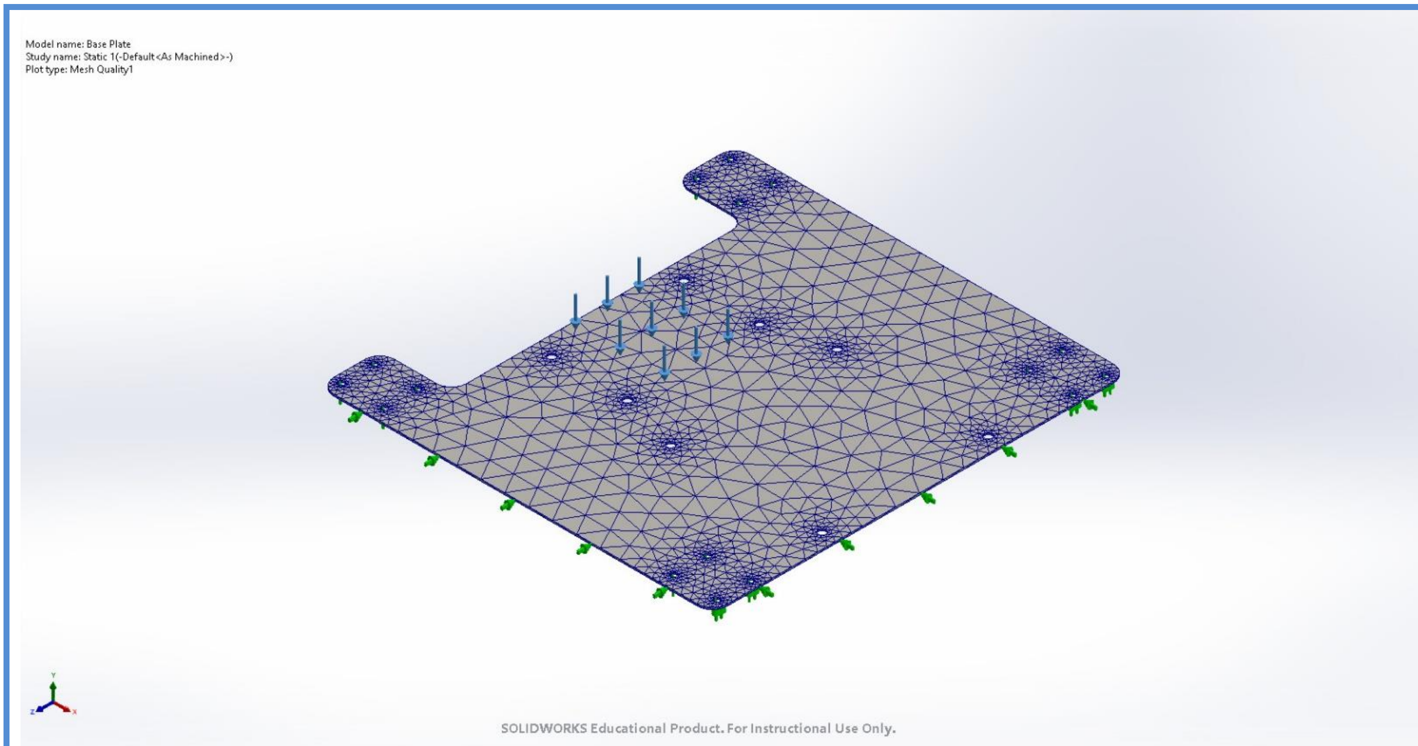
Mesh type	Solid Mesh
Mesher Used:	Blended curvature-based mesh
Jacobian points for High quality mesh	16 Points
Maximum element size	0.792133 in
Minimum element size	0.0765367 in
Mesh Quality	High

Mesh information - Details

Total Nodes	17891
Total Elements	8473
Maximum Aspect Ratio	14.271
% of elements with Aspect Ratio < 3	39.1
Percentage of elements with Aspect Ratio > 10	7.1
Percentage of distorted elements	0
Time to complete mesh(hh;mm;ss):	00:00:07
Computer name:	ENG401700

Mesh Quality Plots

Name	Type	Min	Max
Quality1	Mesh	-	-



Base Plate-Static 1-Quality-Quality1

Sensor Details

No Data

Resultant Forces

Reaction forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-6.10352e-05	222.4	6.8903e-05	222.4

Reaction Moments

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	0

Free body forces

Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N	-9.15527e-05	0.000333428	0.000106454	0.000361785

Free body moments

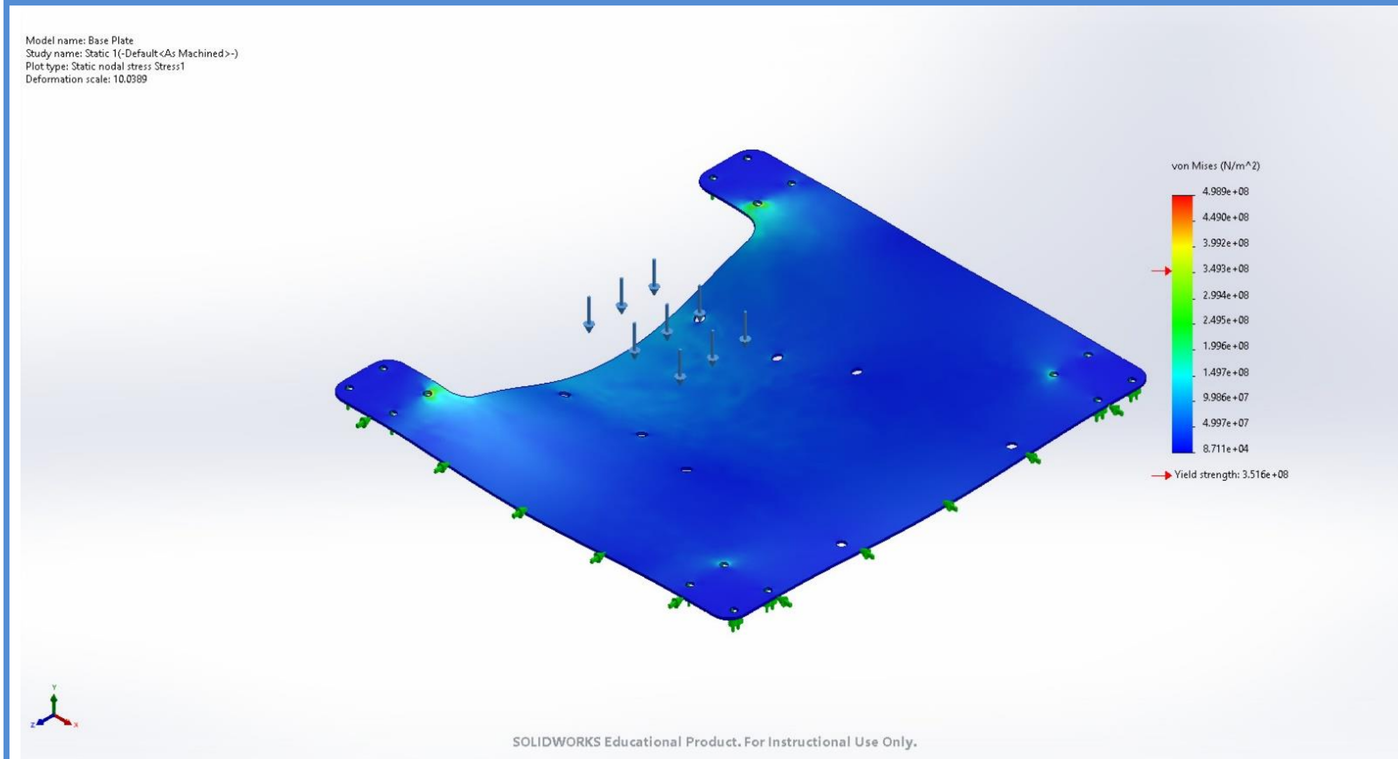
Selection set	Units	Sum X	Sum Y	Sum Z	Resultant
Entire Model	N.m	0	0	0	1e-33

Beams

No Data

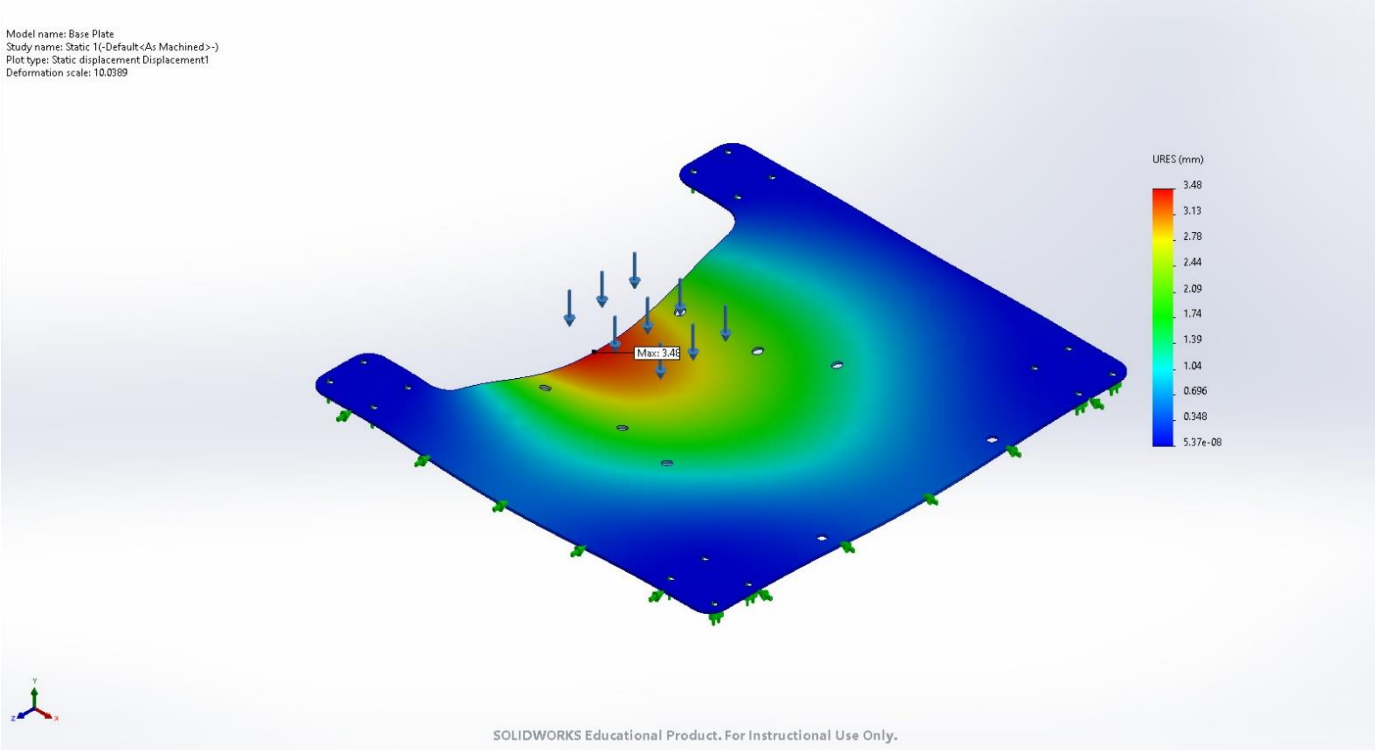
Study Results

Name	Type	Min	Max
Stress1	VON: von Mises Stress	8.711e+04N/m^2 Node: 5313	4.989e+08N/m^2 Node: 14565



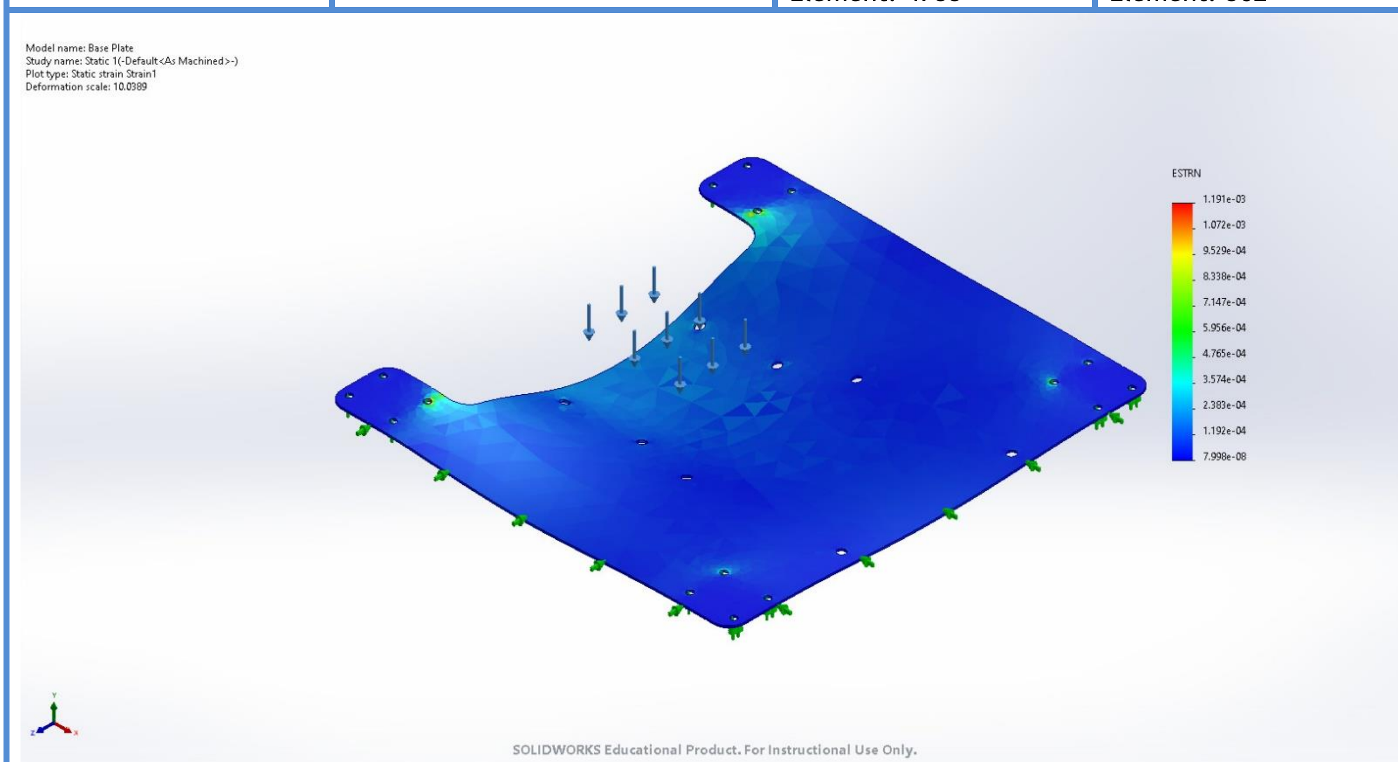
Base Plate-Static 1-Stress-Stress1

Name	Type	Min	Max
Displacement1	URES: Resultant Displacement	5.37e-08mm Node: 3894	3.48mm Node: 134



Base Plate-Static 1-Displacement-Displacement1

Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	7.998e-08 Element: 4788	1.191e-03 Element: 562



Base Plate-Static 1-Strain-Strain1

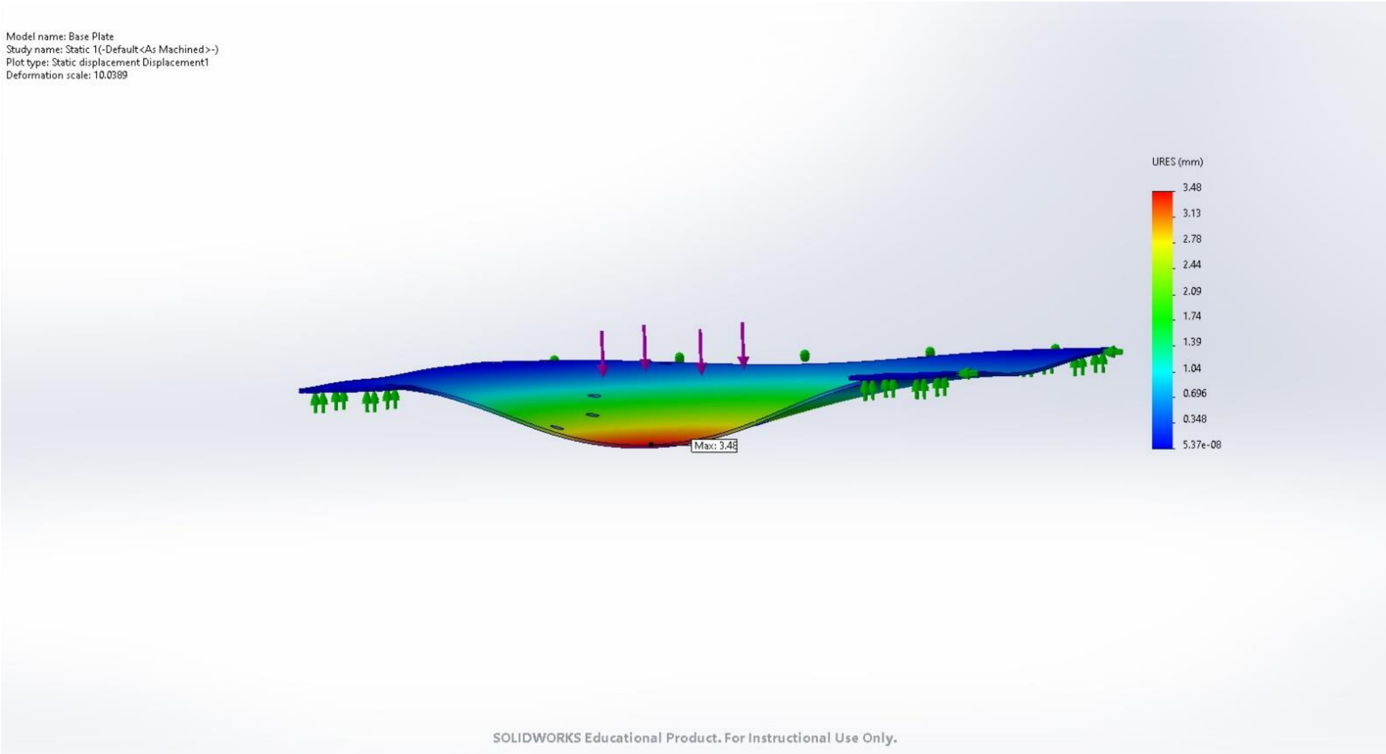


Image-1

Conclusion